

```
"advertised_scheme": "wss",
"advertised_host": "node1.server.com",
"advertised_port": 443
```

- b. Install and start MongoDB.
Use the command:

```
sudo apt install mongodb
```

- c. Install Tendermint as follows:

```
#sudo apt install -y unzip
#wget https://github.com/tendermint/
tendermint/releases/download/v0.22.8/
tendermint_0.22.8_linux_amd64.zip
#unzip tendermint_0.22.8_linux_amd64.zip
#rm tendermint_0.22.8_linux_amd64.zip
#sudo mv tendermint /usr/local/bin
```

- d. Configure Tendermint.
Use the command:

```
tendermint int
```

Step 7: Set up the BigchainDB network.

- a. Most of the things can be done by node operators. However, these node operators, who are also called Members, must share some information with each other so that they can form a network. There is one special Member who helps to coordinate with everyone, called the Coordinator. Each BigchainDB node is identified by its:

- hostname, i.e. the node's DNS sub-domain, such as *bnode.example.com*, or its IP address, such as 10.191.93.111
- Tendermint *pub_key.value*
- Tendermint *node_id*

The Tendermint *pub_key.value* is stored in the file *\$HOME/.tendermint/config/priv_validator.json*.

- b. To get your Tendermint *node_id*, run the following command:

```
tendermint show_node_id
```

- c. Next, create and share the *genesis.json* file. At this point, the coordinator should have received

the data from all the members and should combine them in the file *\$HOME/.tendermint/config/genesis.json*.

- d. The new *genesis.json* file contains the data that describes the network. The key name is the member's moniker; it can be any valid string, but use something human-readable like 'Alice's Node Shop'.

At this point, the coordinator must share the new *genesis.json* file with all members, and each member should have received the *genesis.json* file.

The members must now copy the *genesis.json* file into the local *\$HOME/.tendermint/config* directory. Every member now shares the same *chain_id* and *genesis_time* (used to identify the network) and the same list of validators.

All the members must edit their *\$HOME/.tendermint/config/config.toml* file and make the following changes:

```
moniker = "Name of our node"
create_empty_blocks = false
log_level =
"main:info,state:info,*:error"

persistent_peers = "<Member 1 node
id>@<Member 1 hostname>:26656,\
<Member 2 node id>@<Member 2
hostname>:26656,\
<Member N node id>@<Member N
hostname>:26656,"

send_rate = 102400000
recv_rate = 102400000

recheck = false
```

 **Note:** The list of *persistent_peers* doesn't have to include all nodes in the network.

- e. Start BigchainDB and Tendermint using Monit. To install Monit, use the following command:

```
sudo apt install monit
```

- f. If you installed the BigchainDB Python package as above, you should have the *bigchaindb-monit-config* script in your PATH now. Run the script to build a configuration file for Monit, as follows:

```
bigchaindb-monit-config
```

Run Monit as a daemon, instructing it to wake up every second to check on processes:

```
monit -d 1
```

Monit will run the BigchainDB and Tendermint processes, and restart them when they crash. If the *root bigchaindb_process* crashes, Monit will also restart the Tendermint process.

You can check the status by running *monit status* or *monit summary*. By default, it will collect program logs into the *~/bigchaindb-monit/logs* folder.

- g. Access the Web portal at *http://hostname(ipaddress):9984*.

Your BigchainDB network is now configured, and up and running! **END** 

References

- [1] <https://docs.bigchaindb.com/en/latest/terminology.html>
- [2] <https://www.bigchaindb.com/features/>
- [3] <https://www.bigchaindb.com/>
- [4] <https://www.crunchbase.com/organization/bigchaindb#section-locked-charts>
- [5] <http://docs.bigchaindb.com/projects/server/en/latest/simple-deployment-template/index.html>

By: Anand Nayyar

The author works at Duy Tan University in Vietnam. He loves to explore open source technologies, sensor communications, network security and the Internet of Things.